YIFAN (JOHNNY) YANG

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EDUCATION

Indiana University, Bloomington, *August 2019 — August 2025*
Ph.D. in Mathematics
Advisor: Prof. Louis (Wai-Tong) Fan

University of Arizona *August 2015–May 2018*
B.S. in Mathematics, Magna Cum Laude
Minor in Finance

EMPLOYMENT AND ACADEMIC POSITIONS

Postdoctoral Research Scholars, Arizona State University, AZ *August 2025 — present*

RESEARCH INTERESTS

Probability theory and its applications, stochastic analysis, stochastic partial differential equations, Dirichlet forms, interacting particle systems, stochastic quantization, mathematical biology.

HONORS AND AWARDS

- William P Ziemer Award, Indiana University Bloomington, 2024.
- College of Arts and Sciences Fellowship, Indiana University Bloomington, 2020.
- Excellence in Undergraduate Research Award, Department of Mathematics, University of Arizona, 2018.

PUBLICATIONS AND WORK IN PROGRESS

Work in Progress

1. José Luis Pérez and Johnny (Yifan) Yang. Non-local dirichlet forms and stochastic quantization equations on metric measure spaces. In preparation, 2026+
2. Louis Fan, Johnny (Yifan) Yang, and John Yin. Spatial modeling for virus infection with multiple stages. In preparation, 2026+
3. Louis Fan, Adrián González Casanova, and Yifan (Johnny) Yang. Universality of stochastic fkpp equation. In preparation, 2026+

Preprints

1. Louis Wai-Tong Fan, Zhenyao Sun, and Johnny (Yifan) Yang. Comparison principles for stochastic reaction-diffusion equations on metric measure spaces. *arXiv preprint arXiv:2610.07790*, 2026. 100 pages
2. Adrián González Casanova and Johnny (Yifan) Yang. From individual-based models to general stochastic reaction diffusion equations. *arXiv preprint arXiv:2610.02669*, 2026. 54 pages
3. José Chacón, Adrián González Casanova, Imanol Nuñez, Rafael Peña Miller, José Luis Pérez, and Johnny (Yifan) Yang. Dormancy stabilizes non-transitive competitive dynamics. *arXiv preprint arXiv:2608.17179*, 2026. 20 pages
4. Hongyi Chen and Yifan (Johnny) Yang. Wick renormalized parabolic stochastic quantization equations on rough metric measure spaces. *arXiv preprint arXiv:2605.05442*, 2026. 71 pages

Published

1. Qiantong Liang, Johnny (Yifan) Yang, Wai-Tong Louis Fan, and Wing-Cheong Lo. Patch formation driven by stochastic effects of interaction between viruses and defective interfering particles. *PLOS Computational Biology*, 19(10), 2023
2. Wai-Tong Louis Fan, Yifan Johnny Yang, and Chaojie Yuan. Constrained langevin approximation for the togashi-kaneko model of autocatalytic reactions. *Mathematical Biosciences and Engineering*, 20(3):4322–4352, 2023

OTHER RESEARCH ACTIVITIES

Collaboration with John Yin (University of Wisconsin—Madison) under National Science Foundation (NSF) funding DMS-2348164, Fall 2024 — present.

- We aim to develop a spatial stochastic model for virus growth with mutations on two-dimensional domains.

Extinctions in multi-species stochastic systems, Indiana University, Fall 2023 — present
Collaboration with Louis Fan and John Yin

- We aim to study the extinction phenomenon in multi-species biological systems caused by stochastic effects.

Summer school Stochastic Quantization, Simons Laufer Mathematical Sciences Institute, Summer 2024.

PRESENTATIONS

Talks

- Singular SPDEs on Rough Spaces, Aarhus Summer School in Analysis and Probability, Aarhus University, July 2026.
- Stochastic Quantization Equations on Metric Measure Spaces in the Da Prato–Debussche Regime, Probability and SPDEs Conference, Ohio State University, June 2026.
- Φ^4 on Rough Spaces: Constructing a Quantum Field Measure Beyond Euclidean Geometry, Postdoc seminar, Arizona State University, February 2026.
- Spatial stochastic modeling, CME seminar, Biodesign, Arizona State University, February 2026.
- Spatial stochastic models for population dynamics, Symposium of Probability and Stochastic Processes, Universidad de Sonora, November 2025.
- SPDEs on metric measure spaces, SPDEs Workshop at OSU, June 2025.
- SPDEs on spatial dimensions $d \in [1, 2 + \epsilon)$, Analysis Seminar, University of Illinois Chicago, March 2025.
- SPDEs on spatial dimensions $d \in [1, 2 + \epsilon)$, Probability Seminar, Arizona State University, February 2025.
- Spatial stochastic modeling for population dynamics, Population Genetics Seminar, University of Chicago, February 2025.
- Patch formation driven by stochastic effects, SIAM Conference on Mathematics of Data Science (MDS24), October 2024.
- Stability of Markov Process and Foster-Lyapunov Type Criteria Demonstration via Togashi-Kaneko Model, Graduate Probability Seminar, Indiana University Bloomington, September 2023.
- On Extinction Probability of Stochastic PDE on Metric Graphs, Graduate Student Probability Conference, University of Wisconsin–Madison, September 2022.

Poster Presentations

- Singular SPDE on metric measure spaces, Heat Kernels and Stochastic Analysis, Aarhus University, June–July 2026.
- Constructive QFT on rough spaces, Seminar on Stochastic Processes, Union College, March 2026.
- SPDEs on metric measure spaces, Seminar on Stochastic Processes, Indiana University Bloomington, March 2025.
- SPDE on metric measure spaces, Recent Progress in Stochastic Analysis and its Applications, Loyola University Chicago, July 2024.
- On Extinction Probability of Stochastic PDE on Metric Graphs, Seminar on Stochastic Processes, University of Arizona, April 2023.
- On Extinction Probability of Stochastic FKPP on Metric Graphs, Seminar on Stochastic Processes, Lehigh University, April 2022.

PROFESSIONAL ACTIVITIES

Organizer, Graduate Probability Seminar at Indiana University Bloomington, Fall 2023 — Spring 2024.

TEACHING EXPERIENCE

Teaching

- STP 421: Probability, Fall 2026, Arizona State University.
- MAT 275: Modern Differential Equations, Spring 2026, Arizona State University.
- MAT 243: Discrete Mathematical Structures, Fall 2025, Arizona State University.

Mentoring, Indiana University, Bloomington

- Co-mentored an undergraduate woman for "Emerging Scholars Research Experience for Undergraduate Women," a year-long program at IU (August 2022–May 2023).
- Co-mentored an undergraduate for the Summer REU at Indiana University — Bloomington (Summer 2023).